MACAULAY DURATION

Learning Outcome

- define, calculate, and interpret Macaulay duration.

The prior lesson established Macaulay duration as a holding period for a bond that balances reinvestment and price risk. We now turn to its calculation.

The most straightforward way to understand the Macaulay duration of a traditional fixed-rate bond is that it is the weighted average of the *time to receipt* of the bond's cash flows, where the weights of each cash flow in the calculation are each cash flow's share of the bond's full price (i.e., present value). Before we introduce the general equation, let's take a specific example to demonstrate the calculation of Macaulay duration: BRWA's 10-year, 6.2% (annual) coupon eurobond used in the previous lessons.

Exhibit 4 illustrates the calculation of Macaulay duration at issuance, assuming the bond was issued at par, settled on 15 October 2025, and matures on 15 October 2035. Because this is a corporate bond, we assume a 30/360 day-count basis.

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Exhibit 4:

Macaulay Duration of BRWA Corp. 10-Year, 6.2% Annual Eurobond at Issuance

Coupon	6.20%
Coupon frequency per year	1
Price per 100 par value	100
Yield-to-maturity	6.20%
Settlement date	15 October 2025
Maturity	15 October 2035
Years to maturity	10

(1)	(2)	(3)	(4)	(5)	(6)
				(4) ÷ Sum of present values	(1) × (5)
Period	**Time to receipt**	**Cash flow**	**Present value**	**Weight**	**Time to receipt × weight**
1	1	6.2	5.8380	0.0584	0.0584
2	2	6.2	5.4972	0.0550	0.1099
3	3	6.2	5.1763	0.0518	0.1553
4	4	6.2	4.8741	0.0487	0.1950
5	5	6.2	4.5895	0.0459	0.2295
6	6	6.2	4.3216	0.0432	0.2593
7	7	6.2	4.0693	0.0407	0.2849
8	8	6.2	3.8317	0.0383	0.3065
9	9	6.2	3.6080	0.0361	0.3247
10	10	106.2	58.1942	0.5819	5.8194
			100.0000	1.0000	7.7429

- Columns 1 and 2 show the number of periods and the time to receipt of the cash flow, respectively. If the calculation is done at issuance or on a coupon date, they are equal.

- The fifth column is the weight, or each cash flow's share of total *PV*. Notice that the final payment, $PMT_N + FV_N$, of 106.2 accounts for 58% of the bond's *PV*.
- The sixth column is the time to receipt of the cash flow (Column 2) multiplied by the weight (Column 5). The sum of this column is the Macaulay duration, 7.7429. This is close to Hightest Capital's investment horizon of eight years from the prior lesson, which means that Hightest has balanced its reinvestment and price risk.

Macaulay duration is often quoted as an annualized statistic. Since this bond pays coupons annually, no adjustment is needed. If the bond were a semiannual coupon bond, we would divide the Macaulay duration by 2 (two semiannual periods per year) to obtain an annualized Macaulay duration. When quoted as an annualized statistic, the units for Macaulay duration is years.

Notice how 7.7429 years is less than 10, the time-to-maturity of the bond. This is because Macaulay duration is a present value–weighted average of the time to receipt of the cash flows, while time-to-maturity is the time to receipt of the final cash flow.

The general calculation of Macaulay duration, *MacDur*, that also accounts for partial coupon periods if the calculation is done between coupon dates is shown in Equation 2.

$$\text{MacDur} = \left\{ \begin{array}{l} (1 - t/T)\left[\dfrac{\dfrac{PMT}{(1+r)^{1-t/T}}}{PV^{Full}}\right] + (2 - t/T)\left[\dfrac{\dfrac{PMT}{(1+r)^{2-t/T}}}{PV^{Full}}\right] + \cdots + \\ (N - t/T)\left[\dfrac{\dfrac{PMT + FV}{(1+r)^{N-t/T}}}{PV^{Full}}\right] \end{array} \right\}, \quad 2$$

where

t is the number of days from the last coupon payment to the settlement date;

T is the number of days in the coupon period;

t/*T* is the fraction of the coupon period that has passed since the last payment;

PMT is the coupon payment per period;

FV is the future value paid at maturity, or the par value of the bond;

r is the yield-to-maturity per period; and

N is the number of evenly spaced periods to maturity as of the beginning of the current period.

EXAMPLE 5

Macaulay Duration between Coupon Dates

Now let's compute the Macaulay duration for the BRWA bond *between* coupon dates —specifically, on 11 December 2025, 57 days after issuance. The only difference from the prior example is that we are 57 days in the future, so rather than *t*/*T* = 0 like at issuance, *t*/*T* = 57/360, and each cash flow's time to receipt is 57 days earlier.

Macaulay Duration of BRWA Corp. 10-Year, 6.2% Annual Eurobond Settling 57 Days after Issuance

Coupon	6.20%
Coupon frequency per year	1
Price per 100 par value	100
Yield-to-maturity	6.20%

Years to maturity					10
(1)	(2)	(3)	(4)	(5)	(6)
				(4) ÷ Sum of present values	(1) × (5)
Period	**Time to receipt**	**Cash flow**	**Present value**	**Weight**	**Time to receipt × weight**
1	0.8417	6.2	5.8939	0.0584	0.0491
2	1.8417	6.2	5.5498	0.0550	0.1012
3	2.8417	6.2	5.2258	0.0518	0.1471
4	3.8417	6.2	4.9207	0.0487	0.1872
5	4.8417	6.2	4.6335	0.0459	0.2222
6	5.8417	6.2	4.3630	0.0432	0.2525
7	6.8417	6.2	4.1082	0.0407	0.2784
8	7.8417	6.2	3.8684	0.0383	0.3005
9	8.8417	6.2	3.6426	0.0361	0.3190
10	9.8417	106.2	58.7511	0.5819	5.7273
			100.9570	1.0000	7.5845

Comparing this example to the prior example, we see that the time to receipt of each flow is lower, reflecting the fact that we are 57 days later, which increases the present value of each cash flow. However, the present value weight of each cash flow remains the same. The Macaulay duration (7.5845) is lower than in the prior example (7.7429), reflecting the lower times to receipt.

This example demonstrates an important property of Macaulay duration: it is not static; it falls as time elapses and the time to receipt of each cash flow falls.

Besides a spreadsheet, Macaulay duration also can be calculated using the DURATION function in Microsoft Excel or Google Sheets:

=DURATION(*Settlement*, *Maturity*, *Coupon*, *Yld*, *Frequency*, [*Basis*])

where

Settlement is the date of settlement, in DATE form

Maturity is the maturity date, in DATE form

Coupon is the annual coupon rate

Yld is the annualized yield-to-maturity

Frequency is the periodicity, typically the number of coupon payments per year; it is 1 for annual, 2 for semiannual, 4 for quarterly

[*Basis*] is the day-count convention; 0 or omitted is 30/360, 1 is act/act, 2 is act/360, 3 is act/365

Finally, another approach to calculating Macaulay duration is to use a closed-form equation derived using calculus and algebra, shown as Equation 3:

$$\text{MacDur} = \left\{ \frac{1+r}{r} - \frac{1+r+[N \times (c-r)]}{c \times \left[(1+r)^N - 1\right] + r} \right\} - \frac{t}{T} \qquad 3$$

where

r is the yield-to-maturity per period;

N is the number of evenly spaced periods to maturity as of the beginning of the current period;

t is the number of days from the last coupon payment to the settlement date, and

T is the number of days in the coupon period.

Question Set

1. Consider a bond that has three years remaining to maturity and a coupon of 4% paid semiannually and is priced at 100. The bond's annualized Macaulay duration is *closest* to:

A. 2.801 years.
B. 2.857 years.
C. 3.000 years.

Hide Solution

Solution:

B is correct.

Period	Time to receipt	Cash flow	Present value	Weight	Time to receipt × weight
1	1	2	1.961	0.0196	0.01961
2	2	2	1.922	0.0192	0.03845
3	3	2	1.885	0.0189	0.05654
4	4	2	1.848	0.0185	0.07391
5	5	2	1.811	0.0181	0.09057
6	6	102	90.573	0.9057	5.43438
			100.00000	1.00000	5.71346
Annualized Macaulay duration (years)					2.857

2. An investor with a three-year investment horizon purchases a bond with four years remaining to maturity, a coupon of 1% paid semiannually, and a yield-to-maturity of 0.90%. With respect to interest rate risk, the investor is *primarily* concerned about:

A. rising interest rates.
B. falling interest rates.
C. neither rising nor falling interest rates.

Hide Solution

Solution:

A is correct. The Macaulay duration of the bond is 3.9312 years.

Period	Time to Receipt	Cash Flow	PV	Weight	Time to Receipt × Weight
1.0000	1.0000	0.5000	0.4978	0.0050	0.0050
2.0000	2.0000	0.5000	0.4955	0.0049	0.0099
3.0000	3.0000	0.5000	0.4933	0.0049	0.0147
4.0000	4.0000	0.5000	0.4911	0.0049	0.0196
5.0000	5.0000	0.5000	0.4889	0.0049	0.0243
6.0000	6.0000	0.5000	0.4867	0.0048	0.0291
7.0000	7.0000	0.5000	0.4845	0.0048	0.0338
8.0000	8.0000	100.5000	96.9542	0.9658	7.7260
			100.3920	1.0000	7.8624
			Annualized MacDur (years)		3.9312

concern is rising interest rates.

3. Consider a bond that has five years remaining to maturity, a coupon of 0.5% paid annually, and a yield-to-maturity of –0.20%. The bond's annualized Macaulay duration is *closest* to:

A. 2.4758.
B. 4.9515.
C. 5.3130.

Hide Solution

Solution:

B is correct.

Period	Time to Receipt	CF	PV	Weight	Time to Receipt × Weight
1.0000	1.0000	0.5000	0.5010	0.0048	0.0048
2.0000	2.0000	0.5000	0.5020	0.0048	0.0097
3.0000	3.0000	0.5000	0.5030	0.0049	0.0146
4.0000	4.0000	0.5000	0.5040	0.0049	0.0195
5.0000	5.0000	100.5000	101.5111	0.9806	4.9029
			103.5211	1.0000	4.9515

4. Consider a bond that has five years remaining to maturity, a coupon of 0.5% paid annually, and a yield-to-maturity of –0.20%. Assume it is 91 days into the first coupon period and a 30/360 basis. The bond's annualized Macaulay duration is *closest* to:

A. 4.2371.
B. 4.5054.
C. 4.6987.

Hide Solution

Solution:

C is correct.

Period	Time to Receipt	Cash Flow	PV	Weight	Time to Receipt × Weight
1.0000	0.7472	0.5000	0.5007	0.0048	0.0036
2.0000	1.7472	0.5000	0.5018	0.0048	0.0085
3.0000	2.7472	0.5000	0.5028	0.0049	0.0133
4.0000	3.7472	0.5000	0.5038	0.0049	0.0182
5.0000	4.7472	100.5000	101.4597	0.9806	4.6550
			103.4687	1.0000	4.6987

Note that a straightforward way to calculate Macaulay duration using a spreadsheet between coupon periods is to set the first cash flow's time to receipt as $1 - t/T$ (here = 1 – 91/360) and subsequent times to receipt as the prior plus 1.

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MO Example 5, last column states (6) = (1) * (5); it should be (6) = (2) * (5).

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